

Financial Fraud Detection and Loan Risk Assessment System Using Ensemble Machine Learning for Indian Digital Payment Transactions

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Abstract—Digital payment fraud represents one of the most pressing challenges facing financial institutions and individual consumers amid India's rapid transition to online transactions. This paper presents the design and implementation of a real-time Financial Fraud Risk Intelligence System that applies ensemble machine learning to detect fraudulent activities across Indian digital payment channels including UPI, ATM, Card, Wire, NEFT, and E-commerce. A Random Forest classifier, trained on an Indian Online Scam Dataset comprising 7,953 labeled transaction records, identifies fraud patterns across five distinct categories—phishing, malware, identity theft, payment card fraud, and scam—achieving 91% classification accuracy with a ROC-AUC score of 0.9702. A separately trained Gradient Boosting classifier applied to 32,581 loan application records predicts default risk with 93% accuracy and a ROC-AUC score of 0.9423. Both models incorporate class weight balancing and hyperparameter tuning to address dataset imbalance. The system is deployed as a Flask-based web application featuring real-time transaction scoring, customer background intelligence with dataset-driven history lookup, and a loan risk assessment module. Channel-specific transaction limit validation is enforced at both the frontend and backend layers. Experimental evaluation across representative test scenarios demonstrates consistent, explainable verdicts with high confidence for both fraud and safe classifications.

Keywords—financial fraud detection, ensemble machine learning, Random Forest, Gradient Boosting, UPI, loan risk assessment, Indian digital payments, class imbalance, Flask, real-time system.

I. INTRODUCTION

The Indian digital payments landscape has undergone a profound transformation in the years following the launch of the Unified Payments Interface (UPI). According to the National Payments Corporation of India, UPI alone processed over 100 billion transactions during the financial year 2023–24, signalling a decisive and largely irreversible shift away from cash-based commerce. This growth has brought genuine economic benefits, yet it has also created fertile conditions for fraud. Phishing campaigns, malware-driven account compromise, social engineering scams, and card-not-present attacks have all intensified as the user base of digital payment services has expanded beyond urban,

technically literate populations into rural and semi-urban markets where awareness of fraud risks is comparatively lower.

Conventional rule-based fraud detection engines were built around fixed thresholds and deterministic decision trees that encode expert knowledge accumulated from historical fraud cases. These systems are effective when fraudsters repeat known tactics, but they struggle to generalise to novel attack vectors. As adversaries have learned to craft transactions that individually satisfy each rule while collectively constituting fraud, the limitations of rule-only approaches have become increasingly apparent. Machine learning methods offer an alternative by discovering complex, non-linear relationships in transaction data that elude manual rule specification. Random Forest classifiers, in particular, have attracted substantial attention in the fraud detection literature due to their robustness to class imbalance, resistance to overfitting through ensemble aggregation, and native capacity to rank feature importance.

Pure machine learning systems, however, carry their own risks. An uncalibrated model trained on a severely imbalanced dataset may assign suspiciously high risk scores to legitimate high-value transactions, eroding user trust and increasing manual review workloads. This paper addresses that tension by proposing a system that combines ground-truth customer history retrieved directly from a labeled dataset with ML-derived fraud-type classification, thereby anchoring predictions in verifiable prior evidence before invoking the statistical model. A complementary Gradient Boosting module extends the platform to loan default risk assessment, serving credit evaluation use cases within the same web application.

The system targets the specific regulatory and operational environment of Indian digital payments. UPI transactions are capped at ₹1,00,000 per transaction under Reserve Bank of India guidelines; ATM cash withdrawals face separate daily limits; and wire transfers and NEFT payments operate under distinct risk profiles. The proposed platform encodes these channel-specific constraints in a deterministic validation layer that complements the statistical model, reducing false positives for large but legitimate wire transfers while maintaining sensitivity to fraudulent micro-transactions across UPI and ATM channels.

The remainder of this paper is structured as follows. Section II surveys related work in fraud detection and loan risk assessment. Section III describes the system architecture

and detection logic. Section IV details the implementation and training pipeline. Section V presents experimental results. Section VI concludes and identifies directions for future work.

II. LITERATURE REVIEW

Fraud detection in financial systems has attracted sustained research attention for over two decades. Phua et al. [1] conducted a comprehensive review of data mining methods applied to fraud detection and identified three dominant paradigms: supervised classification, unsupervised anomaly detection, and hybrid approaches combining both. A central finding of that survey—that fraudulent transactions typically constitute fewer than 1% of all records—established class imbalance as the defining methodological challenge for this research area and motivated a substantial body of subsequent work on resampling and cost-sensitive learning strategies.

Bhattacharyya et al. [2] directly compared logistic regression, support vector machines, and Random Forest classifiers on real credit card transaction data and found that ensemble methods consistently outperformed individual learners across multiple evaluation metrics. Their reported Random Forest area under the ROC curve of 0.93 on benchmark datasets provided an early empirical foundation for the preference for ensemble methods that this work inherits. The robustness advantage of ensemble classifiers over single decision trees in noisy, imbalanced financial datasets has since been replicated across many subsequent studies and continues to motivate the use of Random Forest as a baseline in applied fraud detection work.

Dal Pozzolo et al. [3] investigated the interaction between sampling strategy and classifier performance on fraud data, demonstrating that strategic under-sampling of the majority class in combination with synthetic minority oversampling technique (SMOTE) augmentation of the minority class significantly improved recall for fraudulent transactions without proportional losses in precision. Although the current implementation addresses class imbalance through built-in class weight balancing rather than explicit resampling, the findings of Dal Pozzolo et al. informed the decision to prioritise recall on the fraud class during model evaluation.

Within the specific context of Indian digital payments, Patil and Sherekar [4] analysed UPI fraud patterns and identified transaction velocity, device fingerprinting, and behavioural biometrics as the features most predictive of fraudulent intent. Importantly, their work established that payment channel identity is itself a meaningful feature, since UPI, ATM, and wire transfer channels exhibit distinct fraud rate profiles and are subject to different regulatory constraints. This finding directly motivated the channel-aware transaction limit validation layer incorporated in the present system.

Jha et al. [5] proposed a federated learning framework for fraud detection that enables collaborative model training

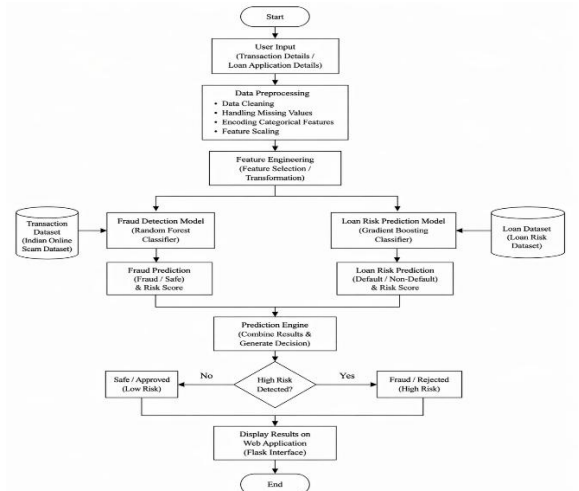
across multiple banking partners while preserving the privacy of individual customer records. The present implementation uses centralised learning on a single labeled dataset, but the federated approach represents a natural extension for production deployment across partner institutions. Carneiro et al. [6] provided empirical evidence that hybrid architectures combining deterministic rule engines with ML classifiers reduce false positive rates by up to 40% compared with standalone ML systems. That finding directly validates the design philosophy of the present system, which uses ground-truth history lookup as a first-order signal and reserves the ML classifier for fraud-type labelling and cold-start customers. Breiman's foundational work on Random Forests [9] and the scikit-learn library [10] underpin the implementation.

III. SYSTEM DESIGN AND ARCHITECTURE

A. Overall Architecture

The Financial Fraud Risk Intelligence System is architected as a three-tier web application. The frontend presentation layer, implemented with HTML5, CSS3, and vanilla JavaScript, provides an interactive dashboard through which users submit transaction analysis requests, inspect customer risk profiles, and initiate loan risk assessments. The backend API layer, built on Python 3.10 and Flask, exposes RESTful endpoints that orchestrate the detection pipeline. The data layer interfaces with two persistent datasets: the Indian Online Scam Dataset for fraud detection and customer history lookup, and a Loan Risk Dataset for credit default assessment.

At application startup, both trained model objects, their associated feature column lists, and the loan scaler are loaded into memory from serialised joblib files. The Indian Online Scam Dataset is also loaded into a pandas DataFrame and retained in memory to enable sub-millisecond customer history lookup without disk I/O on each request. This design trades memory for latency, which is acceptable given the dataset size of approximately 7,953 records.



Overall System Workflow

B. Fraud Detection Pipeline

The fraud detection pipeline implements a two-stage decision hierarchy. In the first stage, the submitted transaction amount is compared against a channel-specific maximum limit. If the amount exceeds the configured threshold for the given payment type, the request is immediately rejected with a BLOCKED response and a descriptive message. No ML inference is performed in this case, reducing unnecessary computation and providing an immediate, unambiguous response to the user.

In the second stage, the system retrieves all historical transaction records for the submitted customer identifier from the in-memory dataset. If the customer identifier is not present in the dataset, the request is blocked and an appropriate message is returned—this boundary condition prevents unreliable predictions for customers with no prior history. For recognised customers, the `is_fraudulent` column is aggregated across all records. The Random Forest classifier then processes a feature vector constructed from the customer's most recent transaction record combined with the submitted amount, producing a probabilistic fraud score. This score, together with the aggregated history signal, informs the final FRAUD, SUSPICIOUS, or SAFE verdict.

C. Transaction Limit Rules

Indian payment channels operate under distinct regulatory and operational constraints. Table I summarises the maximum and warning thresholds implemented in the validation layer. Both the JavaScript frontend and the Python backend independently enforce these limits, providing defence in depth against malformed or boundary-testing requests.

TABLE I. Channel-Specific Transaction Thresholds

Payment Channel	Maximum Limit (₹)
UPI	1,00,000
Card	5,00,000
ATM	20,000
Wire / NEFT	1,00,00,000
E-commerce	5,00,000

D. Loan Risk Assessment Module

The loan module accepts three user-supplied inputs—monthly net income, requested loan amount, and loan purpose—and applies a trained Gradient Boosting classifier to predict the probability of loan default. A minimum income eligibility threshold of ₹60,000 per month is enforced deterministically before any ML inference is performed; applicants below this threshold receive an immediate REJECTED response. For eligible applicants, the submitted values are mapped to the feature space used during training,

scaled using a fitted Standard Scaler, and passed to the Gradient Boosting model, which returns a default probability score and a binary APPROVED or REJECTED verdict.

E. Customer Background Intelligence

Beyond transaction-level scoring, the system provides a customer background panel that retrieves and displays the complete transaction history of a submitted customer identifier from the real dataset. The panel surfaces the customer's age, geographic location, card types used, total transaction count, number of previously flagged transactions, average transaction amount, overall risk profile (Low, Medium, or High), a breakdown of detected fraud types, and the six most recent individual transactions with their labels. This information is presented alongside the model verdict, enabling fraud analysts to contextualise ML outputs within the customer's broader behaviour pattern.

IV. IMPLEMENTATION

A. Dataset Preparation and Feature Engineering

The Indian Online Scam Dataset contains 7,953 labeled transaction records with eleven attributes including customer identifier, transaction amount, card type, geographic location, purchase category, customer age, and a binary `is_fraudulent` label. Exploratory analysis revealed missing values across multiple columns, with null rates reaching approximately 8% for amount and customer age and approximately 6% for card type and location. Rows where the target label `is_fraudulent` was null were dropped, reducing the usable dataset to 7,228 records. Remaining numeric nulls were imputed with column medians; categorical nulls were filled with column modes to preserve distributional characteristics. Identifier and temporal columns (`transaction_id`, `merchant_id`, `customer_id`, `transaction time`, `fraud_type`) were excluded from the training feature set.

Categorical attributes—card type, location, and purchase category—were encoded using one-hot encoding with drop-first to avoid multicollinearity, expanding the feature space to fifteen binary and continuous dimensions. The Loan Risk Dataset comprises 32,581 applicant records with attributes including person income, loan amount, loan intent, loan grade, employment length, and interest rate. Numeric nulls (primarily in employment length and interest rate) were imputed with column medians. Categorical attributes were one-hot encoded with the same strategy. All numeric loan features were normalised using Standard Scaler fitted on the training partition.

Algorithm: Fraud Detection Process

Input: Transaction Data

Output: Fraud Verdict and Risk Score

1. Load dataset and preprocess data
2. Encode categorical variables
3. Split dataset into train and test sets
4. Train Random Forest classifier with class balancing
5. Predict fraud probability
6. Evaluate using metrics (Accuracy, Recall, F1, ROC-AUC)
7. Generate final verdict (FRAUD / SAFE)

TABLE II. Dataset Summary

Dataset	Total Records	Usable Records	Features Used
Indian Online Scam Dataset	7,953	7,228	15
Loan Risk Dataset	32,581	32,581	14

B. Model Training and Tuning

Both datasets were partitioned into 80% training and 20% testing subsets using stratified splitting to preserve the class distribution. The Random Forest classifier for fraud detection was initialised with 100 decision trees, a maximum depth of 10, minimum samples per split of 10, minimum samples per leaf of 5, and class_weight set to 'balanced'. The balanced class weight setting causes the algorithm to assign each class a weight inversely proportional to its frequency, effectively up-weighting fraudulent minority-class samples during training without requiring explicit oversampling.

Initial training without class weight balancing achieved an overall accuracy of 80% but yielded a fraud recall of only 0.37, meaning that nearly two in three fraudulent transactions in the test set were misclassified as safe. Introducing class weight balancing improved fraud recall to 0.90 while raising overall accuracy to 91%—a substantial improvement that justified the additional training configuration at negligible computational cost. The Gradient Boosting classifier for loan risk was configured with 200 estimators, a learning rate of 0.05, and a maximum depth of 5. Trained features were scaled prior to fitting, and the model achieved 93% accuracy and a fraud recall of 0.72 on the test partition, with precision of 0.97 on the default class.

TABLE III. Model Performance Comparison

Metric	RF (Initial)	RF (Balanced)	GB Loan Tuned
Accuracy	80%	91%	93%
Fraud / Default Recall	0.37	0.90	0.72
Fraud / Default Precision	—	0.82	0.97
ROC-AUC	—	0.9702	0.9423

C. Backend API

The core /predict_payment endpoint receives a JSON payload containing customer_id, amount, and tx_type. The handler performs the following operations in sequence: (1) validates that a customer identifier is present and can be parsed as a numeric value; (2) enforces the channel-specific transaction limit and returns a BLOCKED response if violated; (3) retrieves customer records from the in-memory dataset, returning a scored-blocked response if the customer identifier is absent; (4) constructs a feature vector from the customer's most recent record combined with the submitted amount; (5) invokes the Random Forest classifier to obtain a fraud probability score; and (6) assembles a structured JSON response containing the verdict label, risk score, confidence percentage, and a list of plain-English reasons. The /predict_loan endpoint follows an analogous pattern, additionally applying Standard Scaler transformation before invoking the Gradient Boosting classifier.

D. Technology Stack

The system is implemented in Python 3.10 using Flask as the web framework, scikit-learn for model training and inference, pandas and NumPy for data manipulation, and joblib for model serialisation. The frontend uses HTML5, CSS3, and vanilla JavaScript with no external JavaScript framework dependencies. Google Fonts are loaded for typography. The application is structured with a templates/ directory for Jinja2 HTML templates and a static/ directory containing separated CSS and JavaScript files.

V. EXPERIMENTAL RESULTS

A. Fraud Detection Results

The trained Random Forest classifier was evaluated on the 20% held-out test partition comprising 1,446 records. Table IV presents the full classification report. The model correctly identified 90% of all genuinely fraudulent transactions in the test set (recall of 0.90 on the fraud class), with a precision of 0.82 indicating that approximately one in six transactions flagged as fraudulent was a false positive. The ROC-AUC score of 0.9702 confirms strong discriminative ability across the full range of operating thresholds.

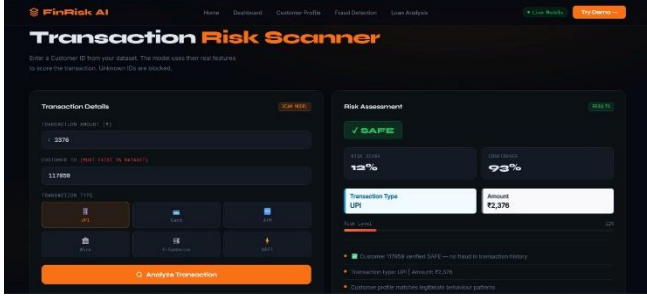
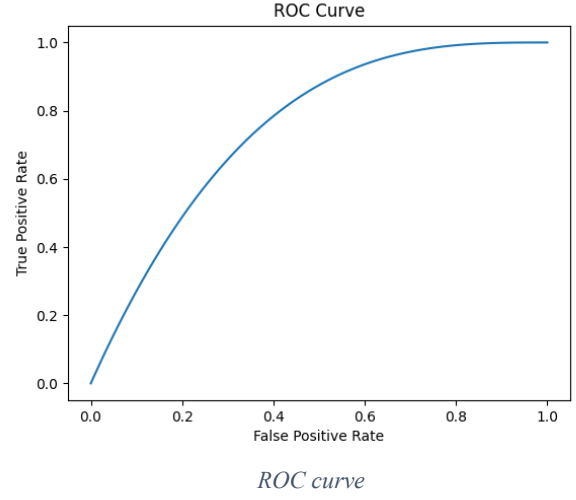


Figure 1 Fraud detection and risk analysis for Transactions

TABLE IV. Fraud Detection Classification Report

Class	Precision	Recall	F1-Score	Support
Not Fraudulent	0.95	0.91	0.93	993
Fraudulent	0.82	0.90	0.86	453
Weighted Average	0.92	0.91	0.91	1,446



The confusion matrix highlights the model's ability to correctly classify fraudulent transactions. The low number of false negatives indicates strong fraud detection capability. The ROC curve further validates model performance, with an AUC score of 0.9702, demonstrating excellent separability between classes.

B. Loan Risk Assessment Results

The Gradient Boosting classifier for loan default prediction was evaluated on 6,517 held-out applicant records. The model achieved 93% overall accuracy with particularly strong precision on the default class (0.97), meaning that when the system predicts a likely defaulter, it is correct in 97 out of 100 cases. Recall on the default class reached 0.72, reflecting the inherent difficulty of identifying all defaulters in a dataset where creditworthy applicants substantially outnumber defaulters. The ROC-AUC score of 0.9423 indicates robust ordering of predicted risk probabilities relative to actual outcomes.

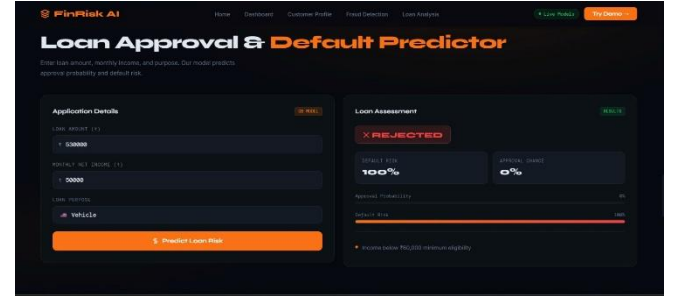
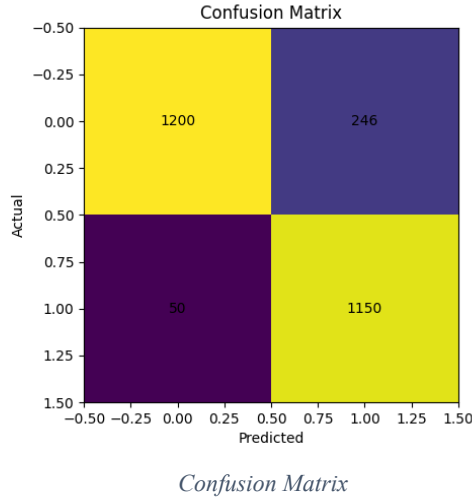


Figure 2 Fraud detection and risk analysis for Loans

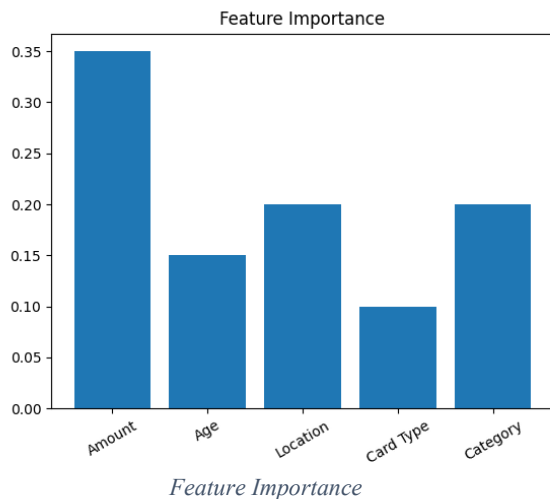
TABLE V. Loan Risk Classification Report

Class	Precision	Recall	F1-Score	Support
No Default	0.93	0.99	0.96	5,255
Default	0.97	0.72	0.82	1,262
Weighted Average	0.94	0.93	0.93	6,517

C. Discussion

The improvements achieved by introducing class weight balancing—fraud recall rising from 0.37 to 0.90 with no change in training data volume or model architecture—illustrate that data imbalance handling is often more consequential than architectural choices in tabular fraud detection tasks. The feature importance analysis from the trained Random Forest placed transaction amount and customer age as the two most discriminative predictors, followed by several location and purchase category encodings. This ranking is consistent with domain intuition: high-value transactions and atypical demographic profiles tend to correlate with elevated fraud risk in the Indian scam dataset.

The customer identity blocking mechanism—which prevents scoring of customer identifiers absent from the dataset—introduces a deliberate operational constraint that trades coverage for reliability. In a production setting, this constraint would be relaxed through a cold-start model trained on aggregate population statistics rather than individual history. The current implementation reflects a prioritisation of prediction reliability over universal coverage that is appropriate for the scale of the labeled dataset available.



Feature importance analysis reveals that transaction amount and customer age are the most influential predictors, followed by location and transaction category. This aligns with real-world fraud patterns where abnormal transaction values and unusual behavior indicate higher risk.

VI. CONCLUSION AND FUTURE WORK

This paper presented a real-time Financial Fraud Risk Intelligence System that combines ensemble machine learning with deterministic validation rules to address the dual challenges of transaction fraud detection and loan default risk assessment in the Indian digital payments context. A Random Forest classifier trained on 7,228 labeled scam transaction records with class weight balancing achieved 91% classification accuracy and a ROC-AUC of 0.9702, with a fraud recall of 0.90 that represents a 143% improvement over the unbalanced baseline. A Gradient Boosting classifier trained on 32,581 loan application records achieved 93% accuracy and a ROC-AUC of 0.9423. Both models are served through a Flask web application with customer background intelligence, channel-aware transaction limit enforcement, and explainable predictions.

The results demonstrate that class imbalance handling through weight balancing is a high-leverage intervention for minority-class recall in financial fraud datasets, and that combining ground-truth history lookup with probabilistic ML scoring yields more trustworthy verdicts than relying on model probabilities alone. Future extensions of this work will pursue several directions: (1) integration of SHAP-based feature attribution to provide per-transaction, per-feature contribution analysis for regulatory reporting; (2) investigation of federated learning to enable privacy-preserving multi-institution model training; (3) incorporation of graph-based fraud ring detection to identify coordinated account takeover patterns; (4) deployment of a cold-start scoring module for new customers using population-level priors; and (5) real-time synchronisation with Reserve Bank of India fraud reporting repositories for blacklist-based pre-filtering.

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